Final Project

In the final project, you will research selected topics in social network theory[1] and apply your knowledge toward some problem domain. This is an opportunity to explore topics not covered in class or take topics from the classroom to new depths. You will also have the opportunity to share your findings with other students. Undergraduates will work in groups of 2 or 3. Graduate students will work solo.

The topic of the project can be a **survey** of social network theory as applied to a domain of interest, an **application** of these tools to a problem of your choice (including the construction and analysis of network datasets based on raw data), or a **development** of your own research ideas related to the course material. You will be required to submit a proposal of your project approximately mid-semester.

In the **application** paper, you will find and analyze graph data for a domain of interest. Visualize or describe the graphs to highlight important characteristics. And apply social network theory to analyze its structure toward answering a well-defined research question.
In the **novel dataset** paper, you will identify a domain of interest, construct a dataset for that domain, and analyze the resulting data. This often includes the creation of a novel dataset by primary observation of media content (such as recorded sports broadcasts or works of fiction[2]), followed by the transformation of the dataset into one more graphs. Your methodology must include the data gathering and graph creation processes, and the soundness of your methods will be part of the evaluation. Appropriate techniques should be used to analyze the graph, though in less depth than an application paper.
You may also use social network theory to further **develop** your own research ideas. This may take a theoretical or a computational approach, or a combination of the two. If you plan to do this, you should run your idea by the instructor first, but generally any novel research ideas are welcome.

The main deliverable for the project is a 10- to 12-page **paper** (1.15-spaced, 12-point font). You may alternatively use the LaTeX submission guidelines of the Autonomous Agents and Multiagent Systems Conference (https://aamas2025.org/index.php/conference/calls/call-for-papers-main-technical-track/); under this guideline, the page expectation is 7 pages, but will actually enable you to fit more content. You may add additional pages for references and figures if needed.

[1] For the purpose of the project, "social network theory" can be very broadly defined. It would be fine to do any project that makes use of graph theory or network science in a significant way.
[2] Character interactions should use the following media template:
https://docs.google.com/spreadsheets/d/1ZbxDqZq3pAhX_opsBdxao5pmu2N_QGoSlxjS4oJhgfU/edit?usp=sharing

Those considering graduate school are strongly encouraged to consider the application or development papers, and to submit their paper in LaTeX.

In addition to your paper, you will also give an in-class live **presentation** in the closing weeks of the term. Because some presentation dates are well in advance of the due date, your presentation will focus on communicating your research question, your plan and progress, and your interim results.

Consultation

Prior to the proposal due date, a class will be set aside for project consultations. This is an opportunity to meet with the instructor as a group to discuss potential project topics. This is optional but highly recommended.

Proposal 3%

The intention of the proposal is for you to get useful feedback about the feasibility of your proposed topic, your research plan, and potentially additional ideas related to the topic. Your proposal should address the following questions:

- What is your topic and what is the research question that you are trying to answer?
- What aspects of the topic interest you? How do you plan on exploring that aspect? This may incorporate additional theoretical frameworks or simulations and experiments.
- Where do you intend on finding resources that will help you answer your research question? Or what relevant literature have you already identified?

Presentation 20%

Your presentation will give you a chance to share your research with the other students. Your project may be far from completion (especially if your presentation date is earlier in the sequence), but you should have a plan or some progress to show. The presentation will be primarily assessed based on the effectiveness of your communication and the quality of plan and progress.

You should showcase any early results you have made and you may use the opportunity to ask the audience for suggestions and ideas. If your presentation makes use of pre-existing models, you must clearly indicate in the presentation that these components are not your own research. Your delivery of the presentation should be natural; you should aim to “teach” the class about your research, and **you must not make use of scripts or extensive notes**.

Due to the limitations of scheduling, presentations may be moved to an online video format. More information will be available after project groups are finalized.

Paper 32%

The final product will constitute a large part of your final grade and will take the place of the course's final exam. The primary deliverable is a paper that presents your research, written in your own words and with appropriate references. You should submit all supplementary materials that you generated as part of your project (ex: data, code, spreadsheets, etc.) as a zip file. Do not submit data that can be readily downloaded (ex: datasets accessible via URLs). If your data files are too large, you can submit a link to a repository or hosted file instead.

The rubric for evaluating your paper is on the next page, as well as a recommended structure for your report. The structure is offered only as a guideline and can be adapted to suit the needs of the particular topic.

In the comments to the final submission, you should include **1 paragraph that describe progress** that you have made to the paper since the Presentation. In particular, you should highlight any feedback that has been incorporated into the paper, and where that can be found. If you had an earlier presentation slot, then this paragraph can be a very high-level summary; if you had a later presentation, then this paragraph may cover only a few specific points.

Peer Feedback 5%

During presentations, you will be required to submit short reviews of the presentations delivered by your peers. The quality of your feedback will be graded for a total of 5% of your final grade. We will also make the anonymized aggregated feedback available to the presenter.

The total number of reviews submitted will be determined by the number of projects and scheduling of presentations, and so is not yet finalized.

Final Project Rubric

Problem Context The research question is well defined and utilizes concepts from social network theory (broadly defined) throughout the paper. The problem domain, especially the relevant aspects, is clearly described.	**30%**
Quality of Research An appropriate approach is selected and documented. The results are clearly described and analyzed or validated (as appropriate to the problem). Progress since the Presentation is clearly documented in the text comments associated with the submission.	**50%**
Clarity of Communications Overall, ideas are clearly and concisely communicated using correct grammar and professional language.	**20%**

Recommended Sections

- You *do not need* a cover page
- **Abstract** that briefly introduces the topic and summarizes your contributions / results
- **Introduction** that guides the reader to the problem
- **Related Works** survey related literature
- **Background** necessary to describe the problem and your solution; you should describe, in your own words, the social network concepts that you will be using (even if they are described in class)
- **Data** – if your paper constructs a dataset or makes extensive use of an existing one, you should describe it here
- **Model** details how your research question can be modeled by networks
- **Methodology** – if your paper contains experimental / simulation components, this section should contain enough details for another researcher to replicate your work
- **Results** detail your findings, be they experimental or theoretical (in the latter case, you may need to expand this into several sections by theme or category)
- **Discussion** may talk about the nature of your Results (including plausible explanations of unusual findings) and their potential application and extensions; this section is sometimes combined with Conclusion
- **Conclusion** that sums up your findings

Presentation Rubric

Problem Description The presentation clearly lays out the problem domain, using the language of social network theory (broadly defined). The central research question is clearly described and relevant to graphs, social networks, or general network structures.	**30%**
Research Plan & Interim Progress The presentation lays out a plan of how the research question will be investigated. The plan is well thought-out and feasible. Highly related work (if applicable) is described. For presentations before **April 1**, you should show you how have started making progress enacting those plans. For presentations scheduled after, you should also present your interim results.	**40%**
Scientific Communication Skills Overall, ideas are clearly expressed during the presentation using a combination of oral and visual communication skills. NB: Delivery should not use scripts or extensive notes.	**30%**

Important Milestones

Dates in this version of the document is **tentative**.

Group Selection	Feb 1
Project Consultation	Week of Feb 12
Proposal Due	Feb 26
Presentations	Last ~3 Weeks of Classes
Final Submission	April 17

Grievances

Students working in groups are expected to collaborate and divide the work in an equal and fair manner. If there is a breakdown in collaboration within the group, any and all remaining components of the project may be completed individually. If this is the case, please apprise the instructor of this as soon as possible.

Academic Integrity

Your presentation and your paper must be in your own words. Where you use or talk about other peoples' ideas, you must cite appropriate references. You should paraphrase their words, though you may use a small number of quotations (clearly marked out using "quotes"). Presentations and papers which violate academic integrity will receive a zero and be passed along to the Dean's Office as per policy. For further details on academic integrity, see the ODS webpage: https://science.carleton.ca/academic-integrity/.

You must **document all generative AI use** related to the project. Submit your summary or logs of AI usage as appendices or supplemental files to your paper. For the course's generative AI use policy, see the Course Syllabus.